Bogoan Kim, Ph.D.

Contact Information

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Research Interest

My research sits at the intersection of **human-computer interaction (HCI)**, **digital health**, and **inclusive AI**. I design and empirically evaluate AI-augmented systems—spanning mobile applications, immersive (VR) environments, and LLM-powered tools—grounded in multimodal behavioral, physiological, and contextual data. A particular emphasis is placed on preserving human agency in AI-driven processes and on the meaningful integration of AI into real-world workflows across diverse professional and social contexts, informed by participatory design and evidence synthesis.

Positions

Chungbuk National University, Cheongju, Korea — Sep 2025 – Present
- *Assistant Professor, School of Information and Communication Engineering*
- *Affiliated Researcher, Cancer Research Center*
- *Director, CBNU HCI Lab*

KAIST, Daejeon, Korea — Mar 2025 – Aug 2025
- *Jang Young-Sil Postdoctoral Fellow, College of Engineering*
  - Host Professors: Profs. Hwajung Hong and Uichin Lee

The Pennsylvania State University, PA, USA — Jan 2020 – Jun 2020
- *Visiting Scholar, College of IST*
  - Mentors: Profs. Dongwon Lee and Aiping Xiong

Indiana University, IN, USA — Jun 2019 – Sep 2019
- *Visiting Scholar, Luddy School of Informatics*
  - Mentors: Profs. Patrick C. Shih and Georgia C. Frey

Education

Hanyang University, Seoul, Korea
- *Ph.D. in Artificial Intelligence*
  - Dissertation: BalancePath: AI-VR Framework for Personalized Adaptive System
  - Advisor: Prof. Kyungsik Han

University of Minnesota – Twin Cities, MN, USA
- *B.S. in Statistics (Graduated with honors)*
  - Thesis: Independent Associations of MVPA with Blood Pressure in Hypertensive Adults
  - Thesis Supervisor: Prof. Barbara Kuzmak

Awards & Honors

ACM CHI 2026 Best Paper Honorable Mention — 2026
- Awarded for two publications, [c.10] and [c.12] (Top 5% of submissions)

Jang Young Sil Fellowship — 2025
- National fellowship for outstanding researchers under the MSIT
- Proposal title: Personalized Discomfort Prediction via Multimodal LLMs

NRF of Korea Ph.D. Fellowship — 2023
- Granted 40M KRW (equivalent to 35,000 USD) stipend for 2-year Ph.D. studies
- Selected among highly competitive STEM applicants nationwide (Top 180 in Open Track)
- Proposal title: AI+VR-based QoLT Systems for Social Independence of Autistic Individuals

SIGCHI Travel Grant for UIST 2023 Doctoral Symposium — 2023

Gary Marsden Travel Award for CHI 2023 — 2023

SK Creative Challenge Finalist — 2022

IITP Best Global Research Team Award — 2021
- Selected as one of 15 teams out of 382 (Top 3.9%)
- Proposal title: Human Factors in Fake News Detection and Echo Chamber Mitigation

IITP High-Potential Individuals Global Training Fellowship 2020
- Support for research @ Pennsylvania State University

Indiana University Global Talent Attraction Program (GTAP) Fellowship 2019
- Support for research @ Indiana University

Honors on the Dean's List 2011, 2012, 2015

Global Excellence Scholarship (merit-based; 4,210 USD) 2011

Special Recognitions for Outstanding Reviews
- CHI 2024, CHI 2025, IMWUT 2025, CHI 2026

Domestic Conference Awards
- Grand Paper Award: KDMS 2024
- Best Paper Award: KDMS 2024
- Best Presentation Award: KCC 2024

PUBLICATIONS

Conference Proceedings (⋆: NRF top-tier CS conferences; *: equal contribution)

[c.13] When Special Education Meets LLMs: Investigating the use of LLM-based Conversational AI by Special Education Teachers for Autism in China
J Wu, D Choi, B Kim, and H Hong
DIS 2026 (*ACM International Conference on Designing Interactive Systems*)

Best Paper Honorable Mention Award (Top 5%)

[c.12] QuerySwitch: Supporting the Design Process by Balancing Vagueness through Large Language Models
⋆ M Kim, B Kim, and K Han
CHI 2026 (*ACM CHI Conference on Human Factors in Computing Systems*)
Links: PDF

[c.11] LAPS: Automating Hypothesis-Driven Statistical Analysis of Public Survey Using Large Language Models
⋆ {J Kim*, D Jeong*}, B Son, H Kim, B Kim, and K Han
CHI 2026 (*ACM CHI Conference on Human Factors in Computing Systems*)
Links: PDF

Best Paper Honorable Mention Award (Top 5%)

[c.10] "I Choose to Live, for Life Itself": Understanding Agency of Home-Based Care Patients Through Information Practices and Relational Dynamics in Care Networks
⋆ {S-I Kim*, J Park*}, B Kim, and H Hong
CHI 2026 (*ACM CHI Conference on Human Factors in Computing Systems*)
Links: PDF

[c.9] Understanding Human-Multi-Agent Team Formation for Creative Work
⋆ H Lim, D Choi, S Nam, B Kim, and H Hong
CHI 2026 (*ACM CHI Conference on Human Factors in Computing Systems*)
Links: PDF

[c.8] "I Don't Know Why I Should Use This App": Holistic Analysis on Engagement Challenges in Mental Health
⋆ {S Jin*, B Kim*}, and K Han
CHI 2025 (*ACM CHI Conference on Human Factors in Computing Systems*)
Links: PDF Media

[c.7] Narrating Routines through Game Dynamics: Impact of a Routine Management App for Autistic Individuals
⋆ B Kim, D Jeong, H Hong, and K Han
CHI 2024 (*ACM CHI Conference on Human Factors in Computing Systems*)
Links: PDF 10-min Talk System Teaser

[c.6] V-DAT: Supporting Self-Awareness for Autistic People from Multimodal VR Sensor Data
⋆ B Kim, D Jeong, JG Kim, H Hong, and K Han
UIST 2023 (*ACM Symposium on User Interface Software and Technology*)
Links: PDF 10-min Talk System Teaser

[c.5] Supporting Independence of Autistic Adults through Mobile and Virtual Reality Technologies
⋆ B Kim (Doctoral Symposium)
UIST 2023 (*ACM Symposium on User Interface Software and Technology*)
Doctoral Symposium
Links: PDF

[c.4] RoutineAid: Externalizing Key Design Elements to Support Daily Routines of Individuals with Autism
⋆ B Kim, S-I Kim, S Park, HJ Yoo, H Hong, and K Han
CHI 2023 (*ACM CHI Conference on Human Factors in Computing Systems*)
Links: PDF 10-min Talk System Teaser

[c.3] VISTA: User-Centered VR Training System for Effectively Deriving Characteristics of Autistic Individuals
⋆ B Kim, D Jeong, M Jeong, T Noh, S-I Kim, T Kim, S-Y Jang, HJ Yoo, JG Kim, H Hong, and K Han
VRST 2022 (*The ACM Symposium on Virtual Reality Software and Technology*)
Links: PDF 10-min Talk System Teaser Media

[c.2] ChamberBreaker: Mitigating the Echo Chamber Effect through a Gamified Inoculation System
⋆ Y Jeon, B Kim, A Xiong, D Lee, and K Han
CSCW 2021 (*ACM Conference on Computer-Supported Cooperative Work and Social Computing*)
Links: PDF Media

[c.1] Photos Don't Have Me, But How Do You Know Me?: Analyzing and Predicting Users on Instagram
K Han, Y Jo, Y Jeon, B Kim, J Song, and S-W Kim
UMAP 2018 (*ACM Conference on User Modeling, Adaptation and Personalization*)
Links: PDF

Journal Articles (*: equal contribution)

[j.11] VR-ACT: Transforming Clinical Evidence of ACT through VR for Subthreshold Depression
B Kim, D Jeong, YS Choi, H Kim, and K Han
Springer Virtual Reality *(SCIE; IF 5.0; JCR **11.2%**; Published 2026.03)*
Links: PDF

[j.10] Mobile and VR Technologies for Social Independence in Autistic Individuals: A Scoping Review
B Kim, D Kang
Journal of Korean Institute of Intelligent Systems *(KCI; Published 2026.03)*

[j.9] Enhancing Mindfulness-Based Cognitive Therapy in a Virtual Reality: A Prospective Interventional Study
{B Kim*, D Jeong*}, YS Choi, Y Choi, H Kim, and K Han
Scientific Reports *(SCIE; IF 3.9; JCR **18.0%**; Published 2025.09)*
Links: PDF

[j.8] Development of a Personalized Early Prediction Model for Cybersickness in Virtual Reality
YS Choi, D Jeong, B Kim, and K Han
KIISE Transactions on Computing Practices *(KCI; Published 2025.06)*

[j.7] Promoting Self-Efficacy of Individuals with Autism in Practicing Social Skills in the Workplace Using Virtual Reality and Physiological Sensors: Mixed Methods Study
S-I Kim, S-Y Jang, T Kim, B Kim, D Jeong, T Noh, M Jeong, HJ Yoo, K Han, H Hong, and JG Kim
JMIR Formative Research *(SCIE; IF 2.20, Published 2024.01)*
Links: PDF Video

[j.6] A Study on Mobile Game Design to Support Daily Routines of Individuals with Autism
B Kim, S-I Kim, S Park, HJ Yoo, H Hong, and K Han
Journal of the HCI Society of Korea *(KCI; Published 2022.05)*

[j.5] Usability Assessment of a Mobile Health Intervention for Increasing Physical Activity in Autistic Adults
DH Lee, GC Frey, A Min, B Kim, DJ Cothran, S Bellini, K Han, and PC Shih
Health Informatics Journal *(SCIE; IF 2.3; Published 2020.09)*
Links: PDF

[j.4] PuzzleWalk: A Theory-Driven Iterative Design of a Mobile Game for Promoting Physical Activity in Autistic Adults
B Kim, DH Lee, A Min, S Paik, GC Frey, S Bellini, K Han, and PC Shih
PLoS ONE *(SCIE; IF 2.9; Published 2020.09)*
Links: PDF

[j.3] No More One Liners: Bringing Context into Emoji Recommendations
J-G Kim, T Gong, B Kim, J Park, W Kim, E Huang, K Han, J Kim, JG Ko, and S-J Lee
ACM Transactions on Social Computing 2020
Links: PDF

[j.2] A Systematic Review on Fake News Research through the Lens of News Creation and Consumption: Research Efforts, Challenges, and Future Directions
B Kim, A Xiong, D Lee, and K Han
PLoS ONE (SCIE, IF = 2.9, Published 2020.09)
Links: PDF

[j.1] ATM: The Effectiveness of Self-Regulation on Time Management through a Smartphone Application
B Kim, S-W Lee, H Hong, and K Han
IEEE Access (SCIE, IF = 3.4, Published 2019.07)
Links: PDF

Posters and Workshops

[p.9] Agreement Between Smartphone- and Research-grade Accelerometer-measured Step Counts in Autistic Adults in Free-living Conditions
L McNulty, **B Kim**, AE Bodde, ML Mauriello, BC Helsel, and D Lee
ACSM 2026 (*American College of Sports Medicine*)

[p.8] Early Prediction of Cybersickness in Virtual Reality Using an LLM for Multimodal Time Series Data
YS Choi, D Jeong, **B Kim**, and K Han
UbiComp 2024 (*ACM Conference on Interactive, Mobile, Wearable and Ubiquitous Technologies*)

[p.7] Objectively Measured Physical Activity and Sedentary Time in Autistic Adults
D Lee, PC Shih, **B Kim**, K Han, A Min, D Cothran, S Bellini, and GC Frey
ACSM 2020 (*American College of Sports Medicine*)

[p.6] A Study on the Text-Based Characteristics of Autistic Individuals through Social Media Data Analysis
J Han, **B Kim**, and K Han
ICNGC 2020 (*The International Conference on Next Generation Computing*)

[p.5] FashionQ: An Interactive Tool for Analyzing Fashion Style Trend with Quantitative Criteria
Y Jeon, S Jin, **B Kim**, and K Han
CHI 2020 (*ACM Conference on Human Factors in Computing Systems*)

[p.4] Ecological Momentary Assessment of Physical Activity and Anxiety in Autistic Adults
D Lee, D Cothran, S Bellini, PC Shih, K Han, **B Kim**, A Min, S Paik, and GC Frey
INSAR 2020 (*International Society for Autism Research*)

[p.3] Bringing Context into Emoji Recommendations
J-G Kim, T Gong, **B Kim**, J Park, E Huang, W Kim, K Han, J Kim, JG Ko, and S-J Lee
MobiSys 2019 (*ACM Conference on Mobile Systems, Applications, and Services*)

[p.2] Iterative Design and Usability of Gamified Application for Adults with Autism Spectrum Disorder
S Paik, **B Kim**, and K Han
ICNGC 2019 (*The International Conference on Next Generation Computing*)

[p.1] Leveraging Mobile Technology to Support Goal Setting and Strategies of College Students
B Kim, S Rhim, K Han, and S-W Lee
UbiComp 2018 (*ACM Conference on Interactive, Mobile, Wearable and Ubiquitous Technologies*)

Patents & Software

Granted Patents

- VR-based Interactive Social Skills Training System for Autistic Individuals — Mar 2026
  Registration Number: **KR10-2938949**
- A Mobile System for Routine Management in Autistic Individuals — Oct 2024
  Registration Number: **KR10-2729263**

Filed Patents

- A Multimodal LLM Capturing Inter-modal Dynamics for Personalized VR Sickness Prediction — Jun 2025
  Application Number: **KR10-2025-0100532**
- VR Data Analysis Tool to Support Self-Awareness for Autistic Individuals — Aug 2023
  Application Number: **KR10-2023-0116955**

Registered Software

- VR-based Dashboard for Clinical Support and Patient Monitoring in Depression Care — Oct 2024
  Registration Number: **C-2024-034698**

Teaching Experience

Faculty Instructor (*: courses taught in English)

- [GR] Special Topics in Deep Learning* (8836552-01) — Spring 2026
- [UG] Information Technology Programming (0914006-10) — Spring 2026
- [UG] Gaesin Frontier II (0281087-12) — Spring 2026
- [UG] Capstone Design (5108102-03) — Spring 2026
- [UG] Advanced Computer Programming (5108113-01) — Spring 2026
- [UG] Intelligent System (5108115-01) — Fall 2025
- [UG] Object-Oriented Programming (5108124-01) — Fall 2025
- [UG] Fundamental Computer Programming (0914002-18) — Fall 2025

Instructor / Co-lecturer (*: courses taught in English)

- Computational Social Science* — Spring 2025
- Human-Computer Interaction* — Fall 2024
- Human-Computer Interaction* — Fall 2023
- Introduction to Software Design* — Spring 2023
- Creative Software Design* — Fall 2022
- Introduction to Software Design* — Spring 2022
- Creative Software Design* — Fall 2021
- Data Structures and Practice* — Spring 2021
- AI Intensive Class — Fall 2019
- Data Structures and Practice* — Spring 2019

Teaching Assistant

- Algorithms — Fall 2020
- Introduction to AI Programming Language — Spring 2020
- The Basics of Data Analytics — Fall 2018
- The Basics of Data Analytics — Spring 2018

Invited Talks

Conference Invited Talk, Dajeon, Korea — Apr 2026

- Host: Korean Society of Veterinary Science Spring Conference 2026
- Session: The Next-Generation Research Paradigm in the Era of AI Transformation (AX)
- Topic: Designing Agency-Centered Interactions in Digital Health

Lab Guest Speaker Session, Seoul, Korea — Aug 2025

- Host: MINDS Lab @ Chung-Ang University
- Topic: Designing Sociotechnical Systems for People in Need

Europe-Korea Conference 2025 Seminar, Seoul, Korea — Aug 2025

- Host: College of Electrical & Computer Engineering @ Chungbuk National University
- Topic: The Role of HCI Researchers in the Development of Digital Therapeutics

Lab Guest Speaker Session, Daejeon, Korea — May 2025
- Host: DxD Lab @ KAIST
- Topic: AI-Infused Software Systems for Solving Real-World Challenges

Omnibus Lecture Series on Computational Social Science, Seoul, Korea — Mar 2025
- Host: Department of Data Science @ Hanyang University
- Topic: User-Centered Digital Healthcare

Computational Social Science Colloquium, Seoul, Korea — May 2024
- Host: Center for Computational Social Science @ Hanyang University
- Topic: Sociotechnical Systems for People Who Are in Need

Top Conference Session Invited Talk, Hongcheon, Korea — Jan 2024
- Host: HCI Korea 2024
- Topic: Impact of a Gamified Routine Management App for Autistic Adults

UIST Doctoral Symposium, San Francisco, CA USA — Oct 2023
- Host: ACM UIST 2023
- Topic: AI-VR Framework for Personalized Adaptive System

Top Conference Session Invited Talk, Jeju, Korea — Jun 2023
- Host: Korea Computer Congress (KCC) 2023
- Topic: Externalizing Key Design Elements to Support Daily Routines of Autistic Adults

IITP Global Talent Program Workshop, Seoul, Korea — Dec 2021
- Host: Institute of Information & Communications Technology Planning & Evaluation (IITP)
- Topic: Explainable Fake News Detection with Deep Learning and Human Factors

HCI Korea Doctoral Colloquium, Seoul, Korea — Jan 2021
- Host: HCI Korea 2021
- Topic: Designing Inclusive Mobile Applications for Marginalized Populations

Lab Guest Speaker Session, PA, USA — Feb 2020
- Host: The PIKE Group @ The Pennsylvania State University
- Topic: From Creation to Consumption: Data-Driven Perspectives on Fake News Research

Invited Guest Lecture, Suwon, Korea — Oct 2019
- Host: AI Intensive Class @ Ajou University
- Topic: Designing a Gamified mHealth App for Marginalized Populations

Global Talent Attraction Program Conference, IN, USA — Jun 2019
- Host: Luddy School of Informatics, Computing, and Engineering @ Indiana University
- Topic: Designing a Mobile Game to Promote Physical Activity in Autistic Adults

Global Challenge Program Symposium, Tokyo, Japan — Jun 2018
- Host: Environment and Information Studies @ Keio University
- Topic: Effectiveness of a Smartphone App in Self-Regulating Time Management

Academic Services

Conference Reviewer
- The Conference on Human Factors in Computing Systems (**CHI**) — 2019 - 2026
- The Conference on Interactive, Mobile, Wearable and Ubiquitous Technologies (**IMWUT**) — 2019 - 2025
- The Designing Interactive Systems Conference (**DIS**) — 2025
- The Conference on Computer-Supported Cooperative Work & Social Computing (**CSCW**) — 2019 - 2024
- The Conference on Information and Knowledge Management (**CIKM**) — 2021 - 2023
- The Symposium on Applied Computing (**SAC**) — 2022
- The Symposium on Mixed and Augmented Reality (**ISMAR**) — 2021 - 2022

Journal Reviewer
- npj Digital Medicine — 2026
- ACM Transactions on Accessible Computing — 2026
- JMIR Formative Research — 2025
- The Nature Scientific Reports — 2024
- The Taylor and Francis Interactive Learning Environments — 2023

Student Volunteer

- The Conference on Mobile Systems, Applications, and Services (**MobiSys**) 2019

Academic Mentorship

Mentored Students

- C Lee, Chungbuk National University (Master's Student) Fall 2025 - Present
- S Seo, Chungbuk National University (Undergrad. Student) Fall 2025 - Present
- D Jeong, Chungbuk National University (Undergrad. Student) Fall 2025 - Present
- D Lee, Chungbuk National University (Undergrad. Student) Fall 2025 - Present
- J Kim, Hanyang University (Ph.D. Student) Spring 2025 - Fall 2025
- L McNulty, University of Delaware (Ph.D. Student) Spring 2025 - Fall 2025
- H Lim, KAIST (Ph.D. Student) Spring 2025 - Fall 2025
- S-I Kim, Seoul National University (MD, Ph.D. Student) Fall 2024 - Fall 2025
- J Park, KAIST (Master's Student) Spring 2025 - Fall 2025
- M Kim, Hanyang University (Ph.D. Student) Spring 2024 - Present
- YS Choi, Hanyang University (Master's Student) Fall 2023 - Present
- D Jeong, Hanyang University (Ph.D. Student) Spring 2022 - Present
- S Paik, Ajou University (Master's Student) Spring 2019 - Fall 2020

Press & Interviews

Research Media Coverage

- "인간의 가치를 지키는 'AI 가치 정렬' 연구 성과..." Mar 2026
- "정신건강관리 앱 지속사용 한계..." May 2025
- KAIST 등, 자폐스펙트럼장애 직장 적응 돕는 VR 개발 Jan 2023
- New game can help users identify, avoid online echo chambers Oct 2021

References

Hwajung Hong, *Professor* (Postdoc. Host Professor) hwajung@kaist.ac.kr
Department of Industrial Design, KAIST

Uichin Lee, *Endowed Chair Professor* (Postdoc. Host Professor) uclee@kaist.ac.kr
School of Computing, KAIST

Kyungsik Han, *Professor* (Ph.D. Advisor) kyungsikhan@hanyang.ac.kr
Department of Data Science, Hanyang University

Daehyoung Lee, *Assistant Professor* (Collaborator) dhlee@udel.edu
Department of Health Behavior and Nutrition Sciences, University of Delaware

Jennifer G. Kim, *Assistant Professor* (Collaborator) jennifer.kim@cc.gatech.edu
School of Interactive Computing, Georgia Institute of Technology

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